The $\log^2$ term, and the final finite result in colour-singlet form

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1 Summary

References: "FS n" is Eq. (n) of `Finite_terms_simplified.pdf` (updated with this note); "EC n" is Eq. (n) of `Evolution_terms_corrected.pdf`. Notation as in FS: $a = \frac{k^+}{\vee}$, $L_\vee = \log\frac{\vee}{k^+}$, $L_* = \log\frac{\sqrt{\vee\Lambda}}{k^+}$, $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5 k^+}$.

1. The $\log^2$ term (Sec. 2). It is in the final result, and that is correct for your file as it stands.

- It appears once in `Finite_terms_1.tex`: in the bracket of Group I Row II part 1, $-\frac{67}{9} + \frac{\pi^2}{3} + \log^2(1-a) + \log(1-a)L_\vee$. No other row has a $\log^2$, so nothing cancels it.
- It is the finite part of the dressed one-ρ state $|\Psi^A_{g\rho}\rangle + |\Psi^B_{g\rho}\rangle$. I re-derived it analytically (Sec. 2.2): it comes from the lower limit $\xi \geq \frac{k^+}{\vee}$ of the ξ integral of $|\Psi^B_{g\rho}\rangle$, i.e. from requiring that the parent gluon has momentum $\frac{k^+}{\xi} < \vee$.
- The same limit gives the $\log(1-a)$ in the pole bracket of Row II part 1, i.e. the uncancelled pole of Q1. The $\log^2$ and the Q1 pole come together.
  - If the limit is kept (as in your file), both stay.
  - If it is dropped where it is not needed, both disappear and Row II part 1 leaves only $\frac{67}{9} - \frac{\pi^2}{3}$. "Not needed" means the parts of the integrand that are regular at $\xi \to 0$; this is AABKL's $\int_0^1 d\xi$.
  - One cannot drop the $\log^2$ and keep the rest.
- There is no separate $\log^2$ in the result: together with its partner,

$$-\log^2(1-a) - L_\vee \log(1-a) = -\log\Big(1 - \frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+} - 1\Big),$$

  which is $\approx a\log\frac{1}{a} \to 0$ for $k^+ \ll \vee$.
- It is not a running-coupling term. In the final result it is now written separately as $V_\vee$, next to the Q1 pole, and the running-coupling remainder is the CMW constant $\frac{67}{9} - \frac{\pi^2}{3}$ alone. FS (4.3) and FS (11.1) are updated accordingly.

2. Final result in colour-singlet form (Secs. 3–7). The projectile is in a colour singlet,

$$\langle \rho^a(u)\rho^b(v)\rangle = \delta^{ab}\,\frac{\langle \rho^f(u)\rho^f(v)\rangle}{N_c^2-1} \equiv \delta^{ab}\,\mathcal{P}(u,v).$$

- Every colour factor of the final result FS (11.1) is projected and written with traces of adjoint U, $U^\dagger$, T^e.
- This covers LO, every LO-like term, the four two-ρ rows, and the two-ρ parts of all three-ρ rows (merge by merge).
- The index placements are those valid for a general adjoint U (Sec. 3.2). For $U = U^T$ (your convention $U^{bc} = U^{cb}$) they reduce to your colour factors.
- The three-ρ symmetric parts need $\langle\rho\rho\rho\rangle$ and vanish for a Gaussian projectile. The four-ρ term (Q2) needs $\langle\rho\rho\rho\rho\rangle$.
- The result is Eq. (7.1).

Checks (Sec. 8): every trace form equals the direct contraction of its colour network with random general SU(3) Wilson lines, to $\leq 6.0\times10^{-16}$. The general-U colour operators equal the first-principles colour functions, to $\leq 1.1\times10^{-14}$.

2 The $\log^2$ term

2.1 Where it is

Group I Row II part 1 (the first equation of Group I Row II in `Finite_terms_1.tex`), bracket

$$\Big[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\Big]\Big[\frac{11}{3}+\log\frac{\vee}{k^+}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\Big(1-\frac{k^+}{\vee}\Big)+\log\Big(1-\frac{k^+}{\vee}\Big)\log\frac{\vee}{k^+}. \quad (2.1)$$

With its c.c. it gives FS (4.1). The β_0 term cancels against Row V 2b UV. The rest is

$$\frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO}\ \underbrace{-\frac{\alpha_s}{2\pi}N_c\Big[\log^2(1-a)+L_\vee\log(1-a)\Big]\mathrm{LO}}_{V_\vee}+\text{pole terms}\ \propto\big(2L_\vee+\log(1-a)\big). \quad (2.2)$$

2.2 Where it comes from

`Calculation_main.tex`, Eqs. for $|\Psi^A_{g\rho}\rangle$ and $|\Psi^B_{g\rho}\rangle$ (one ρ, common factor $\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}k^2}$, $P_\epsilon\equiv-\frac{2}{\epsilon}+\log\frac{k^2}{\mu^2_{\overline{MS}}}$):

$$|\Psi^A_{g\rho}\rangle:\quad P_\epsilon\Big[\frac{11}{3}+\log\frac{\Lambda}{k^+}-2\log\frac{\vee}{\Lambda}\Big]+\frac{\pi^2}{3}-\frac{67}{9}+\log^2\frac{\Lambda}{k^+}-\log^2\frac{\vee}{k^+},$$

$$|\Psi^B_{g\rho}\rangle:\quad \int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{\sqrt{\xi}(1-\xi)}\Big[(\xi-2)\log\xi+(\xi+1)\log(1-\xi)-(\xi-2)P_\epsilon\Big]\,|g(\xi k^+)\rangle.$$

In $|\Psi^B\rangle$ the gluon has momentum ξk^+; k^+ is the momentum of its parent. Call the gluon momentum k^+ again, i.e. substitute $k^+\to k^+/\xi$ (Jacobian included, the measure becomes $\frac{d\xi}{\xi(1-\xi)}$). The limits become:

- parent below the cutoff, $\frac{k^+}{\xi}<\vee$: $\quad\xi>\frac{k^+}{\vee}=a$;
- $1-\xi>\frac{\Lambda\xi}{k^+}$: $\quad\xi<1-\delta+O(\delta^2)$, $\delta=\frac{\Lambda}{k^+}$.

So

$$|\Psi^B\rangle\ \to\ \int_a^{1-\delta}\frac{d\xi}{\xi(1-\xi)}\Big[(\xi-2)\log\xi+(\xi+1)\log(1-\xi)-(\xi-2)P_\epsilon\Big]. \quad (2.3)$$

Pole part: $\frac{2-\xi}{\xi(1-\xi)}=\frac{2}{\xi}+\frac{1}{1-\xi}$, so

$$\int_a^{1-\delta}\Big(\frac{2}{\xi}+\frac{1}{1-\xi}\Big)=2L_\vee+\log(1-a)+\log\frac{k^+}{\Lambda}.$$

Finite part, term by term:

integrand	$\int_a^{1-\delta}d\xi$	comes from
$-2\frac{\log\xi}{\xi}$	$L_\vee^2$	soft end $\xi\to0$ (needs the limit a)
$-\frac{\log\xi}{1-\xi}$	$\mathrm{Li}_2(1-a)$	
$\frac{\log(1-\xi)}{\xi}$	$\mathrm{Li}_2(a)-\frac{\pi^2}{6}$	
$2\frac{\log(1-\xi)}{1-\xi}$	$\log^2(1-a)-\log^2\frac{\Lambda}{k^+}$	**the** $\log^2(1-a)$

With $\mathrm{Li}_2(a)+\mathrm{Li}_2(1-a)=\frac{\pi^2}{6}-\log a\log(1-a)$ the sum is $L_\vee^2+L_\vee\log(1-a)+\log^2(1-a)-\log^2\frac{\Lambda}{k^+}$. Adding $|\Psi^A\rangle$:

$$P_\epsilon\Big[\frac{11}{3}+2L_\vee+\log(1-a)-2\log\frac{\vee}{\Lambda}\Big]+\frac{\pi^2}{3}-\frac{67}{9}+\log^2(1-a)+L_\vee\log(1-a). \tag{2.4}$$

This is exactly the bracket of Eq. (2.1) (your $\log\frac{\vee}{k^+}+\log(\frac{\vee}{k^+}-1)=2L_\vee+\log(1-a)$, and the $-2\log\frac{\vee}{\Lambda}$ is the evolution part). Numerically (`log2_origin.py`), $\Psi^A+\Psi^B$ minus the bracket is $\le 1.6\times10^{-6}$ (finite part) and $\le 1.0\times10^{-7}$ (pole part), i.e. $O(\delta\log\delta)$.

2.3 What this means

1. **It is correct for your file**, and it is in the final result. Nothing cancels it:
   - it is the only $\log^2$ in `Finite_terms_1.tex`;
   - it multiplies the LO structure. The only other terms with that structure are Row V 2b UV (no $\log(1-a)$) and Group II Row IV. Group II Row IV, in the plus distribution restricted to $\xi>a$, has a single $\log(1-a)$, which is absorbed in $D_{h/g}$ (FS (5.1)).
2. **Compact form.** Exactly,
$$V_\vee\equiv-\log^2(1-a)-L_\vee\log(1-a)=-\log(1-a)\log\frac{1-a}{a}=-\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big). \tag{2.5}$$
   - It is small unless k^+ is close to $\vee$: $V_\vee/(a\log\frac{1}{a})=1.0028,\ 1.0000,\ 1.0000$ for $a=10^{-2},10^{-4},10^{-6}$.
   - It grows only at the end point $k^+\to\vee$, like $-\log^2(1-a)$.
3. **It comes with the Q1 pole.** The $\log(1-a)$ of the pole, the $\log^2(1-a)$ and the $L_\vee\log(1-a)$ all come from the limit $\xi>a$ applied to the four terms of Eq. (2.3) that are regular at $\xi\to0$: $\frac{1}{1-\xi}$, $\frac{\log\xi}{1-\xi}$, $\frac{\log(1-\xi)}{\xi}$, $\frac{\log(1-\xi)}{1-\xi}$.
   - The terms $\frac{2}{\xi}$ and $-\frac{2\log\xi}{\xi}$ do need the limit: they give $2L_\vee$ and $L_\vee^2$, the rapidity logarithms.
   - If the four regular terms are integrated from $\xi=0$ (as in AABKL, where the virtual term is $\int_0^1 d\xi$), the pole bracket is $\frac{11}{3}+2L_\vee-2\log\frac{\vee}{\Lambda}$ and the finite part is $\frac{\pi^2}{3}-\frac{67}{9}$ only. Numerically: $\le1.6\times10^{-6}$.
   - Then the pole tally of FS Sec. 5 closes ($\frac{11}{6}$, absorbed in $D_{h/g}$), Q1 disappears, and so does $V_\vee$.
4. **So there are two consistent options:**
   (i) keep the limit $\xi\ge\frac{k^+}{\vee}$ everywhere (your file): $V_\vee$ and the Q1 pole are both in the result;
   (ii) apply it only where it is needed (the soft end): neither is.

   Which one is right is the question Q1. The final result below is written for (i), with $V_\vee$ and the Q1 term marked; for (ii) delete both lines.

3 Colour-singlet projection: rules

3.1 Projection and traces

- $\langle\rho^a(u)\rho^b(v)\rangle=\delta^{ab}\mathcal{P}(u,v)$, $\mathcal{P}(u,v)\equiv\frac{\langle\rho^f(u)\rho^f(v)\rangle}{N_c^2-1}$. $\mathcal{P}$ is symmetric (the commutator term is $\propto\langle\rho^c\rangle=0$), so the order of the two charges does not matter.
- After the projection, all colour indices are contracted among Wilson lines and structure constants. The network is written with traces, as in `Evolution_singlet.pdf`:
  - $(T^e)_{bc}=-if^{ebc}$; U is real orthogonal, so $U^T=U^\dagger$;
  - two f's joined by a chain C of Wilson lines: $\sum_d C^{ad}T^d=C^\dagger T^aC$, giving $\mathrm{Tr}\{T^eAT^eB\}$ or $\mathrm{Tr}\{AT^e\}\mathrm{Tr}\{BT^e\}$;
  - $UU^\dagger=1$, $\mathrm{Tr}\,1=N_c^2-1$, $T^eT^e=N_c$, $\mathrm{Tr}\{T^eAT^eA^\dagger\}=N_c\mathrm{Tr}A$.
- After the projection, terms that differ only in the point of one Wilson line are combined into differences, e.g. $\mathrm{Tr}\{(U(x)-U(y))(U^\dagger(x')-U^\dagger(y'))\}$.
- Every trace below is real (for real U; a trace with two T^e is real too). So the c.c. of a Group I or Group III term is: complex-conjugate every number (also the i's of the colour factors), then exchange $w\leftrightarrow w'$ (resp. $w\leftrightarrow y'$), as in FS Sec. 2.1.

3.2 Index placement for a general U

Your files use $U^{bc} = U^{cb}$. For the projection the order matters: for a general U, a misplaced index gives, e.g., $\mathrm{Tr}\{U(p)U(q)\}$ instead of $\mathrm{Tr}\{U^\dagger(p)U(q)\}$. The projections below use:

- **Group I** (Rows II, III, IV, V): your colour factors as written. They agree with first principles for a general U (`Group_I_check.pdf`); re-checked here: $\leq 3.1\times10^{-15}$.
- **LO**: $[U(y')-U(x')]^{ce'}\rho^{e'}(x')\,[U(y)-U(x)]^{ce}\rho^{e}(x)$. Your $U^{ce}(x)$ next to $U^{ec}(y)$ is the same for $U=U^T$.
- **Groups II and III**: the placements of correction M4 (`Groups_II_III_check.pdf`, `fixed23.py`). Every changed Wilson line is a $U(z)$ or $U(y')$:

where	your file	general U
$\mathcal{R}^{ab}(w,z,x)$ (all of Group II Row I, Group III Rows I, II, III(2))	$f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$	$f^{acd}U^{cb}(z)U^{de}(x)\rho^e(x)$
Group II Row I, two-charge bracket	$U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)$	$U^{c'b}(z)U^{ad'}(x')U^{c'e'}(y)$
Group III Row I	$U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')$	$U^{d'b}(z)U^{d'e'}(y)U^{ac'}(w')$
Group III Row II	$U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)$	$U^{c'd'}(y)U^{ab'}(x')U^{c'b}(z)$
Group III Rows III, V, VI, III(2) (conjugate side)	$U^{e'c'}(y')$, $U^{bd'}(z)U^{c'e'}(x')$	$U^{c'e'}(y')$, $U^{d'b}(z)U^{c'e'}(x')$
Group III Rows III(1), VI (amplitude side)	$U^{bc}(z)U^{ad}(y)U^{ce}(x)$	$U^{cb}(z)U^{ad}(y)U^{ce}(x)$
Group II Row I part 2, $\mathcal{O}_{58}$	see Eq. (3.1)	

- With these placements every colour operator equals the first-principles colour function of `fixed23.py` for a general U ($\leq 1.1\times10^{-14}$). For $U=U^T$ it equals your operator ($\leq 2.5\times10^{-14}$). As written, your operators differ for a general U (by 14%–220%).
- $\mathcal{O}_{58}$ with the general placement is the product of the two $\mathcal{R}$'s:

$$\begin{aligned}\mathcal{O}_{58} = \mathcal{R}^{ab}(w',z,x')\,\mathcal{R}^{ab}(w,z,x) = {}& f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x)\\ &- f^{c'bd'}f^{acd}U^{ac'}(w')U^{cb}(z)U^{de}(x)\rho^{d'}(x')\rho^{e}(x)\\ &- f^{ac'd'}f^{cbd}U^{ac}(w)U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')\rho^{d}(x)\\ &+ N_c U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x).\end{aligned} \tag{3.1}$$

3.3 Three and four ρ's

- A three-ρ colour operator is $\mathcal{O}=S(\mathcal{O})+\mathcal{O}^{(2)}$ (FS (9.1)). $\mathcal{O}^{(2)}$ has two ρ's and is projected like any two-ρ term (Sec. 6).
- $S(\mathcal{O})$ needs the three-point function. For a colour singlet, the only invariant tensors are d^{abc} and f^{abc}:

$$\langle S\big(\rho^a(u)\rho^b(v)\rho^c(t)\big)\rangle = \frac{N_c\,d^{abc}}{(N_c^2-4)(N_c^2-1)}\langle d^{efg}\rho^e(u)\rho^f(v)\rho^g(t)\rangle + \frac{f^{abc}}{N_c(N_c^2-1)}\langle f^{efg}\,S\big(\rho^e(u)\rho^f(v)\rho^g(t)\big)\rangle. \tag{3.2}$$

  Both vanish for a Gaussian (McLerran–Venugopalan-type) projectile.
- Four ρ's (FS (10.1), Q2): $\langle\rho\rho\rho\rho\rangle$; for a Gaussian projectile it is the sum of the three pairings, $\mathcal{PP}$. The term is zero if Q2 is answered with the range $(\Lambda,\vee)$.

4 LO and the LO-like terms

$$\boxed{\mathcal{L}[f]\equiv\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{P}(x,x')\,\mathrm{Tr}\big\{\big(U^\dagger(x)-U^\dagger(y)\big)\big(U(x')-U(y')\big)\big\}} \tag{4.1}$$

(x, x': charges; y, y': the measured gluon in the amplitude and its conjugate). $\mathcal{L}_\xi[f]$: the same with $e^{-ik\cdot(y'-y)}\to e^{-ik\cdot(y'-y)/\xi}$. The projected LO colour factor is $\mathrm{Tr}\big\{\big(U(x)-U(y)\big)\big(U^\dagger(x')-U^\dagger(y')\big)\big\}$ (agreement 1.4×10^{-14}).

The LO-like terms of FS (11.1) in singlet form ($X=x-y$, $X'=x'-y'$, $\Delta=y'-y$):

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{RC}} = \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathcal{L}[1], \tag{4.2}$$

$$\left.\frac{d^3N}{d^3k}\right|_{V_\vee} = -\frac{\alpha_s}{2\pi}N_c\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\mathcal{L}[1]\qquad\text{(with Q1)}, \tag{4.3}$$

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{Q1}} = \frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\mathcal{L}\big[\hat{E}_X+\hat{E}_{X'}\big], \tag{4.4}$$

$$\frac{d^3N}{d^3k}\Big|_{L_*} = \frac{\alpha_s N_c}{2\pi}\, L_*\, \mathcal{L}\Big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\Big], \tag{4.5}$$

$$\frac{d^3N}{d^3k}\Big|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}(\xi)\, \mathcal{L}_\xi\big[\ell_\Delta(\mu_F)\big], \tag{4.6}$$

$$\begin{aligned}\frac{d^3N_{\rm lin}}{d^3k} = \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2} 2N_c\xi(1-\xi)\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot\Delta/\xi}\Big[\mathcal{K}(X)\cdot\mathcal{K}(X') - 2\big(\mathcal{K}(X)\cdot\hat\Delta\big)\big(\mathcal{K}(X')\cdot\hat\Delta\big)\Big] \\ \times \mathcal{P}(x,x')\,\mathrm{Tr}\big\{\big(U^\dagger(x)-U^\dagger(y)\big)\big(U(x')-U(y')\big)\big\},\end{aligned} \tag{4.7}$$

with $\hat E_r = \frac{2}{\epsilon} + \log\frac{r^2\mu^2_{\overline{MS}}}{c_0^2}$, $\ell_\Delta = \log\frac{\Delta^2\mu_F^2}{c_0^2}$, $P^{(r)}_{gg,+} = 2N_c[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)]$, $\hat\Delta = \Delta/|\Delta|$. Eq. (4.5) is one of the L_* terms that add up to the remainder of Eq. (7.1).

5 The two-ρ rows

The kernels are those of FS Sec. 7, unchanged; only the colour operator is replaced by $\mathcal{P}(u,v)\,C$, with u,v the two charges.

Group I Row V, 2a (FS (7.6), +c.c.):

$$\begin{aligned}\frac{d^3N_{2a}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5 k^+}\int_{x,y,z,x',w'}\{\dots\}_{\rm V2}\,\mathcal{P}(x',y)\,C_{2a} + \text{c.c.}, \qquad \{\dots\}_{\rm V2}\text{ of FS (7.5)},\\ C_{2a} &= N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(x)\big\}\\ &\quad + \mathrm{Tr}\big\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\big\}\\ &\quad - \mathrm{Tr}\big\{T^e U(x)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}.\end{aligned} \tag{5.1}$$

Group I Row V, 2b finite (FS (7.7), +c.c.):

$$\begin{aligned}\frac{d^3N_{2b,\rm fin}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5 k^+}\int_{x,y,z,x',w'}\Big\{\{\dots\}_{\rm V2}\,\mathcal{P}(x',y)\,C_{2b}\\ &\quad + e^{-ik\cdot(w'-z)}\Big(\frac{11}{6}+2L_*\Big)\frac{\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\;(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}\,\mathcal{P}(x',y)\,C'_{2b}\Big\} + \text{c.c.},\\ C_{2b} &= -N_c\,\mathrm{Tr}\big\{\big(U(x)-U(y)\big)\big(U^\dagger(x')-U^\dagger(w')\big)\big\},\\ C'_{2b} &= N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)\big(U^\dagger(y)-U^\dagger(z)\big)\big\}.\end{aligned} \tag{5.2}$$

Group II Row I part 2 (FS (7.9), no c.c.; c_λ, $c_{\lambda'}$, c_L, c_ℓ of FS (7.9)):

$$\begin{aligned}\frac{d^3N_{\rm I(2)}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5 k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\Big[c_\lambda\lambda + c_{\lambda'}\lambda' + c_L L_* + c_\ell\log(1-a)\\ &\quad - \frac{d_\perp}{4}\frac{1-a}{(z-w)^2(z-w')^2}\Big]\mathcal{P}(x',x)\,C_{58},\\ C_{58} &= N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(w')\big\} + N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\ &\quad - \mathrm{Tr}\big\{T^e U(w)U^\dagger(x')T^e U(z)U^\dagger(w)\big\}\\ &\quad - \mathrm{Tr}\big\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(w')\big\}.\end{aligned} \tag{5.3}$$

Group III Row III part 2 (FS (7.12), +c.c.; $c_{1..5}$, $\mathbb{I}_{1..5}$, s, $\mathcal{G}$, $\mathcal{J}(b)$ of FS Sec. 7.3):

$$\begin{aligned}\frac{d^3N_{\rm III(2)}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\Big\{c_1\mathbb{I}_1 + \frac{c_2}{k^+}\mathbb{I}_2 + c_3\mathbb{I}_3 + \frac{c_4}{k^{+2}}\mathbb{I}_4 + \frac{c_5}{k^+}\mathbb{I}_5\\ &\quad - \frac{s}{k^+ B}\big[\mathcal{G} + L_* + \mathcal{J}(b)\big]\Big\}\mathcal{P}(x',x)\,C_{\rm III(2)} + \text{c.c.},\\ C_{\rm III(2)} &= -N_c\,\mathrm{Tr}\big\{\big(U(x')-U(y')\big)U^\dagger(x)\big\}\\ &\quad + \mathrm{Tr}\big\{T^e U(w)U^\dagger(x')T^e U(x')U^\dagger(z)\big\}\\ &\quad - \mathrm{Tr}\big\{T^e U(w)U^\dagger(y')T^e U(y')U^\dagger(z)\big\}.\end{aligned} \tag{5.4}$$

6 The three-ρ rows: two-ρ parts

For a three-ρ row with charges (u_1,u_2,u_3) and kernel $\mathcal{K}_{\rm row}$, the two-ρ part is

$$\frac{d^3N^{(2)}_{\rm row}}{d^3k}=\frac{1}{(2\pi)^3}\int(\text{phase})\sum_{\text{merges}}\mathcal{K}_{\rm row}\big|_{u_j\to u_i}\,\mathcal{P}(u_i,u_k)\,C^{\rm row}_{u_j\to u_i}\ (+\text{c.c.}),\tag{6.1}$$

where $u_j\to u_i$ is the merged pair (the point u_j is set equal to u_i; the integral runs over the points of the row without u_j), u_k the third charge, and C the projected colour factor of $\mathcal{O}^{(2)}$ for that merge. The merges and kernels are those of FS Sec. 9 (the colour now with the general-U placement). Merges not listed are zero.

Group I Row II. Kernel $\mathcal{K}_{\rm II}$: FS (7.1). Colour $\mathcal{O}_{\rm II}$ with the general-U placement; charges (x',x,y); +c.c.

$$\begin{aligned}\frac{d^3N^{(2)}}{d^3k}=\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\Big\{&\mathcal{K}_{\rm II}\big|_{x\to x'}\,\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\}-i\,N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(w')\big\}\\&+i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y)U^\dagger(w')\big\}\Big]\\&+\mathcal{K}_{\rm II}\big|_{y\to x'}\,\mathcal{P}(x',x)\Big[-i\,N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(w)\big\}+i\,N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\&-i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(x')U^\dagger(w')\big\}\Big]\Big\}+\text{c.c.}\end{aligned}\tag{6.2}$$

Group I Row III. Kernel $\mathcal{K}_{\rm III}$: FS (7.2). Colour $\mathcal{O}_{\rm III}$ with the general-U placement; charges (x',x,y); +c.c.

$$\begin{aligned}\frac{d^3N^{(2)}}{d^3k}=\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\Big\{&\mathcal{K}_{\rm III}\big|_{y\to x}\,\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(x)\big\}\\&+\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\&-\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\\&+\mathcal{K}_{\rm III}\big|_{x\to x'}\,\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(x')\big\}\\&-\tfrac{i}{2}\,\mathrm{Tr}\big\{\big(U(y)-U(z)\big)U^\dagger(w')T^eU(w')U^\dagger(x')T^e\big\}\Big]\\&+\mathcal{K}_{\rm III}\big|_{y\to x'}\,\mathcal{P}(x',x)\Big[-\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\&+\tfrac{i}{2}\,\mathrm{Tr}\big\{\big(U(x')-U(z)\big)U^\dagger(w')T^eU(w')U^\dagger(x)T^e\big\}\\&+\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\Big\}+\text{c.c.}\end{aligned}\tag{6.3}$$

Group I Row V part 1. Kernel $\mathcal{K}_{\rm V1}$: the whole of FS (7.4) before $\mathcal{O}_{\rm V1}$, with its prefactor $\frac{ig^4}{32\pi^5k^+}$ and both phases. Colour $\mathcal{O}_{\rm V1}$ with the general-U placement; charges (x',y,y'); +c.c.

$$\begin{aligned}\frac{d^3N^{(2)}}{d^3k}=\frac{1}{(2\pi)^3}\int\Big\{&\mathcal{K}_{\rm V1}\big|_{y\to x'}\,\mathcal{P}(x',y')\Big[i\,N_c\,\mathrm{Tr}\big\{U(x')U^\dagger(y')\big\}-i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y')U^\dagger(w')\big\}\\&+i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\&-i\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\\&+\mathcal{K}_{\rm V1}\big|_{y'\to x'}\,\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\big\{U(x')U^\dagger(y)\big\}+i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x')T^eU(y)U^\dagger(w')\big\}\\&-i\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\&+i\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\Big\}+\text{c.c.}\end{aligned}\tag{6.4}$$

The $\mathcal{K}_{\rm IV}$ family (Group I Row IV, Group III Rows I, II, Group II Row I part 1). Kernel $p_8\mathcal{K}_{\rm IV}$: FS (7.3), (8.1). Colour $\mathcal{O}_{\rm IV}-\mathcal{O}_{\rm G3I}-\mathcal{O}_{\rm G3II}-\frac12\mathcal{O}_A$ with the general-U placement; charges (x',x,y); +c.c. (for the Group II term: block B).

$$\begin{aligned}\frac{d^3N^{(2)}}{d^3k}=\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\Big\{&p_8\mathcal{K}_{\rm IV}\big|_{y\to x}\,\mathcal{P}(x,x')\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(x')-U(w')\big)U^\dagger(w)\big\}\\&+\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\big\}\\&-\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\Big]\end{aligned}$$

$$
\begin{aligned}
&+ p_8 \mathcal{K}_{\rm IV}\big|_{x\to x'}\,\mathcal{P}(x',y)\Big[\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(x')\big\}+\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(w')\big\}\\
&\qquad-\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(w')U^\dagger(x')T^eU(z)U^\dagger(w')\big\}\Big]\\
&+ p_8 \mathcal{K}_{\rm IV}\big|_{y\to x'}\,\mathcal{P}(x',x)\Big[-\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(w)U^\dagger(x')\big\}-\tfrac{i}{2}N_c\,\mathrm{Tr}\big\{U(x)U^\dagger(x')\big\}\\
&\qquad+\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU(x)U^\dagger(x')T^eU(x')U^\dagger(z)\big\}\\
&\qquad+\tfrac{i}{2}\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(x')T^eU^\dagger(x')U(z)\big\}\Big]\Big\}+\text{c.c.}
\end{aligned}
\tag{6.5}
$$

The $\mathcal{K}_C$ family (Group III Rows III part 1, V, VI). Kernel $\frac{ig^4}{16\pi^5k^+}\mathcal{K}_C$: FS (7.11), (8.3). Colour $\mathcal{O}_{\rm III(1)}+2\mathcal{O}_{\rm V}+2\mathcal{O}_{\rm VI}$ with the general-U placement; charges (x',x,y); +c.c. $(w\leftrightarrow y')$.

$$
\begin{aligned}
\frac{d^3N^{(2)}}{d^3k}=\frac{1}{(2\pi)^3}\int e^{-ik\cdot(y'-w)}\Big\{&\frac{ig^4}{16\pi^5k^+}\mathcal{K}_C\big|_{y\to x}\,\mathcal{P}(x,x')\Big[-i\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(x)T^e\big\}\\
&+i\,\mathrm{Tr}\big\{\big(U(z)-U(w)\big)U^\dagger(y')T^eU(y')U^\dagger(x)T^e\big\}\Big]\\
+\frac{ig^4}{16\pi^5k^+}\mathcal{K}_C\big|_{x\to x'}\,&\mathcal{P}(x',y)\Big[-i\,N_c\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')\big\}\\
&+i\,\mathrm{Tr}\big\{\big(U(x')-U(z)\big)U^\dagger(y')T^eU(y')\big(U^\dagger(y)-U^\dagger(w)\big)T^e\big\}\\
&+i\,\mathrm{Tr}\big\{\big(U(y)-U(w)\big)U^\dagger(x')T^eU(x')U^\dagger(z)T^e\big\}\Big]\\
+\frac{ig^4}{16\pi^5k^+}\mathcal{K}_C\big|_{y\to x'}\,&\mathcal{P}(x',x)\Big[i\,N_c\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(x')\big\}\\
&-i\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(x')T^eU(x')U^\dagger(w)T^e\big\}\\
&-i\,\mathrm{Tr}\big\{\big(U(x)-U(z)\big)U^\dagger(y')T^eU(y')\big(U^\dagger(x')-U^\dagger(w)\big)T^e\big\}\Big]\Big\}+\text{c.c.}
\end{aligned}
\tag{6.6}
$$

7 Final result in colour-singlet form

For hadron production ($\overline{MS}$, $d_\perp=2-\epsilon$, $n_f=0$), with $a=\frac{k^+}{\vee}$ and all L_* terms of the rows collected into the remainder (FS (6.2): their sum is $\mathcal{N}\log\frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}-\mathcal{B})$):

$$
\begin{aligned}
\frac{d^3N_h}{d^3k}&=\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^2}\,D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k}&=\mathcal{L}[1]\big|_{\alpha_s(\mu_F)}\qquad\text{(Eq. (4.1))}\\
&+\mathcal{N}\int_{x,y,z,w,w'}e^{-ik\cdot(w'-w)}\,\mathcal{P}(x,y)\Big[\frac{1}{2}\log\frac{\vee}{\Lambda}\big(\mathcal{J}_s+\mathcal{B}_s\big)+\log\frac{k^+}{\sqrt{\vee\Lambda}}\big(\mathcal{J}_s-\mathcal{B}_s\big)\Big]\qquad\text{(evolution; remainder)}\\
&+\frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathcal{L}[1]\qquad\text{(running coupling, CMW constant)}\\
&-\frac{\alpha_s}{2\pi}N_c\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\mathcal{L}[1]+\frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\mathcal{L}\big[\hat{E}_X+\hat{E}_{X'}\big]\qquad(V_\vee\text{ and Q1})\\
&-\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathcal{L}_\xi\big[\ell_\Delta(\mu_F)\big]+\frac{d^3N_{\rm lin}}{d^3k}\qquad\text{(DGLAP, Eqs. (4.6), (4.7))}\\
&+\Big[(5.1)+(5.2)+(5.3)+(5.4)\Big]_{L_*\to0}\qquad\text{(two-}\rho\text{ rows)}\\
&+\Big[\text{Eq. (6.1) for G1 II, III, V(1), the }\mathcal{K}_{\rm IV}\text{ and }\mathcal{K}_C\text{ families}\Big]_{L_*\to0}\qquad\text{(three-}\rho\text{ rows, two-}\rho\text{ parts)}\\
&+\frac{d^3N_{3\rho,S}}{d^3k}\ \big(\propto\langle\rho\rho\rho\rangle\text{; zero for a Gaussian projectile}\big)+\frac{d^3N_{4\rho}}{d^3k}\ \text{(Q2)}\\
&+O(\alpha_s^3)+O\Big(\frac{\Lambda}{k^+}\Big).
\end{aligned}
\tag{7.1}
$$

- $\mathcal{J}_s$, $\mathcal{B}_s$: the singlet brackets EC (3.4) (JIMWLK⊗LO) and EC (4.3) (BFKL⊗LO). In them x, y are the amplitude and conjugate charges and w, w' the measured gluon (here: x, x' and y, y').

- $L_* \to 0$: the rows are written without their L_* terms, which are all in the remainder $\log \frac{k^+}{\sqrt{\vee\Lambda}}(\mathcal{J}_s - \mathcal{B}_s)$; it vanishes at mid rapidity. Equivalently, keep the L_* terms in the rows (and in Eq. (4.5)) and drop the remainder; not both.
- Group I and Group III terms include their c.c.; Group II terms have none.
- For option (ii) of Sec. 2 delete the $V_\vee$ and Q1 line.
- LO is $O(\alpha_s)$; every other term is $O(\alpha_s^2)$. Nothing is expanded in $\frac{k^+}{\vee}$.

8 Checks

Scripts in `nlo_gluon_paper/finite_singlet/` (they use `../singlet/trace.py`, `../finite_v2/compact.py`, `../groups23/fixed23.py`, `../group1/user1.py`):

check	result
$\log^2$: $\Psi^A + \Psi^B$ (numerical ξ integral) vs. the bracket of Row II part 1, $a = 0.05 \ldots 0.9$ (`log2_origin.py`)	$\leq 1.6 \times 10^{-6}$ / $\leq 1.0 \times 10^{-7}$
the same with the regular terms from $\xi = 0$, vs. $\frac{\pi^2}{3} - \frac{67}{9}$ and no $\log(1-a)$	$\leq 1.6 \times 10^{-6}$
the four closed-form integrals of Sec. 2.2	$\leq 9.7 \times 10^{-25}$
general-U operators (Groups II, III) vs. `fixed23.py`, general U (`gsrc.py`)	$\leq 1.1 \times 10^{-14}$
the same vs. your operators for $U = U^T$	$\leq 2.5 \times 10^{-14}$
Group I operators as written vs. `user1.py` (first principles), general U	$\leq 3.1 \times 10^{-15}$
two-ρ row colours vs. first-principles colour functions	$\leq 3.8 \times 10^{-15}$
two-ρ parts by the pair rule vs. the reordering engine, general U	$\leq 2.4 \times 10^{-16}$
every trace form vs. direct contraction (before / after combining differences)	$\leq 6.0 \times 10^{-16}$ / $\leq 6.0 \times 10^{-16}$
LO trace, Eq. (4.1), vs. the projected LO colour	agreement 1.4×10^{-14}

All colour checks use random SU(3) adjoint Wilson lines with $U^T \neq U$ and three random configurations; the relative differences are with respect to the sum of the moduli of the single terms.